E9 222 Signal Processing in Practice
Assignment 8

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Problem 1: MixUp — Interpolating Data for Robust Augmentation

Ans:
Architecture: SimpleCNN — Conv1(3→32) → BN → ReLU → Pool → Conv2(32→64) → BN → ReLU → Pool → FC1(4096→128) → BN → ReLU → FC2(128→10). Trained with Adam (lr=0.001), Cross-Entropy loss, 20 epochs on CIFAR-10 (normalised to $[-1, 1]$).

Accuracy Summary:

Method	Test Accuracy
1.1 Raw (no augmentation)	0.7392
1.2 All augmentations (HFlip + Crop + ColorJitter + Gaussian Noise)	0.7214
1.3 Mixup $\alpha = 0.1$	0.7334
1.3 Mixup $\alpha = 0.2$	0.7346
1.3 Mixup $\alpha = 0.4$	**0.7556**
1.3 Mixup $\alpha = 1.0$	0.7530

Problem 1.1 — Raw Training:

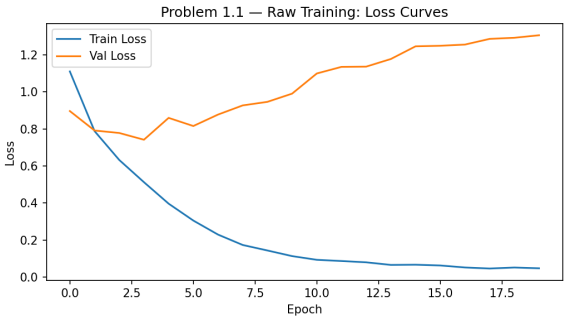


Figure 1: Training and validation loss — raw (no augmentation).

Problem 1.2 — All Augmentations:

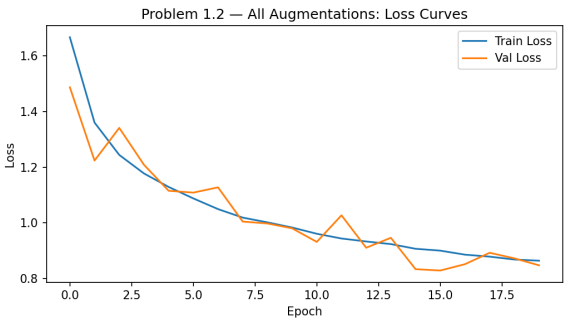
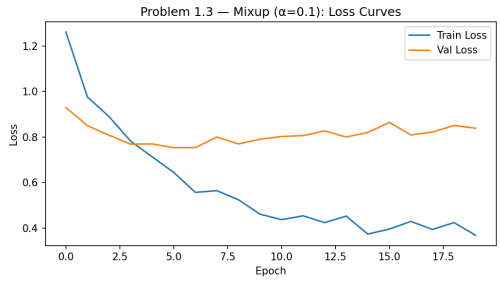
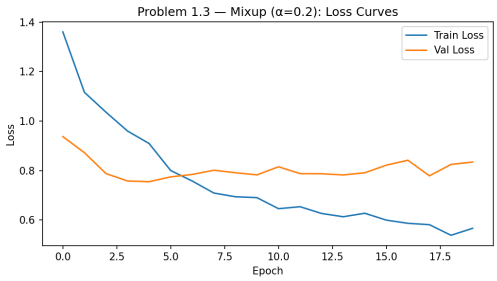


Figure 2: Training and validation loss — all augmentations combined.

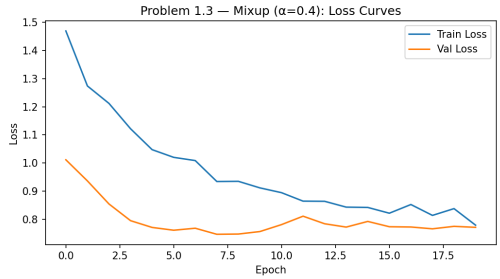
Problem 1.3 — Mixup ($\lambda \sim \mathrm{Beta}(\alpha, \alpha)$):



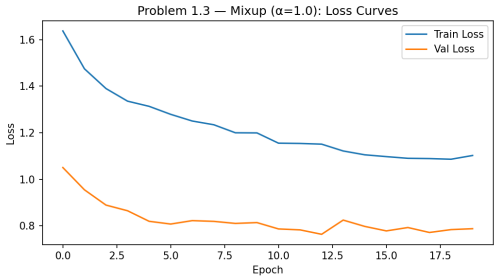
(a) $\alpha = 0.1$



(b) $\alpha = 0.2$



(c) $\alpha = 0.4$



(d) $\alpha = 1.0$

Figure 3: Mixup training and validation loss for different α values. Higher α raises train loss (stronger mixing) but narrows the train/val gap.

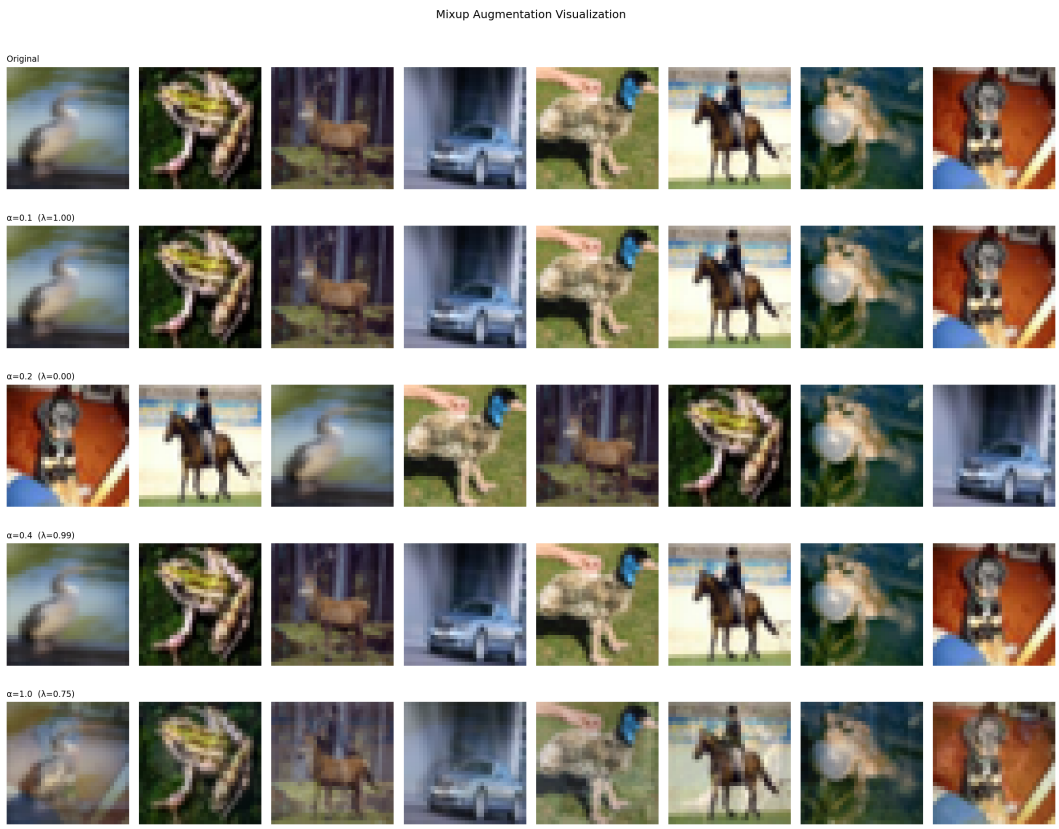


Figure 4: Mixup augmentation visualisation for $\alpha \in \{0.1, 0.2, 0.4, 1.0\}$. Each row shows the same 8 images blended with a random partner; $\alpha = 1.0$ produces the most aggressive blending.

Observations:

- The raw data clearly overfits. The validation curve diverges while the training loss converges. The augmentation reduces overfitting and give a more generalized performance.
- Traditional augmentation (1.2) *hurts* accuracy vs. the raw baseline — aggressive spatial/colour transforms at training time increase noise without providing the interpolation-based regularisation that Mixup offers.
- Mixup consistently *outperforms* the raw baseline. Best accuracy is at $\alpha = 0.4$ (0.7556), a $\sim$1.6 pp gain over raw.
- Small α (0.1, 0.2) keeps λ near 0 or 1, so mixing is mild — train loss stays low but val loss is higher (less regularisation). Larger α (0.4, 1.0)

increases the train/val loss gap but the *val loss is lower*, indicating better generalisation.

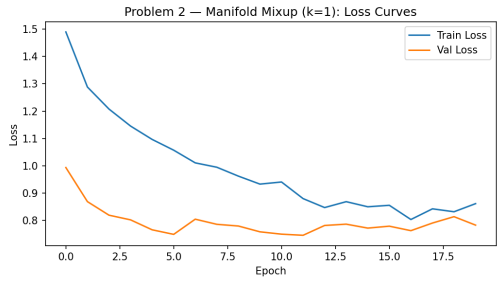
- $\alpha = 1.0$ (Uniform mixing) performs slightly worse than $\alpha = 0.4$, suggesting over-regularisation for this shallow network.

Problem 2: Manifold Mixup — Interpolation in Feature Space

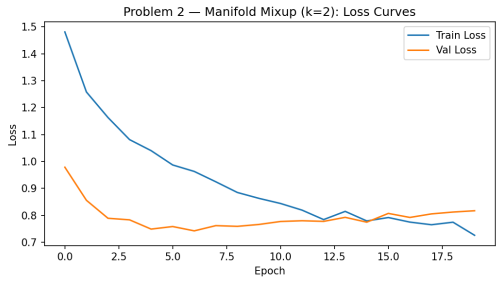
Ans:
Manifold Mixup applies $\tilde{g}_k = \lambda\, g_k(x_i) + (1-\lambda)\, g_k(x_j)$, $\tilde{y} = \lambda y_i + (1-\lambda) y_j$ where g_k is the output of the k-th hidden layer. The same SimpleCNN is used with $\alpha = 0.2$; the final classification layer (FC2) is excluded from mixing.

Layer key: $k = 1$: after Conv1+Pool; $k = 2$: after Conv2+Pool; $k = 3$: after FC1; $k =$ random: uniformly random $k \in \{1, 2, 3\}$ per batch.

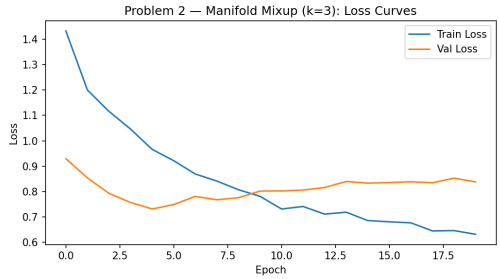
Mixing Layer	**Test Accuracy**
$k = 1$ (after Conv1)	**0.7522**
$k = 2$ (after Conv2)	0.7419
$k = 3$ (after FC1)	0.7429
$k =$ random	0.7399



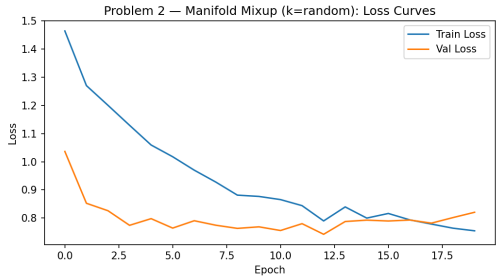
(a) $k = 1$ (after Conv1+Pool)



(b) $k = 2$ (after Conv2+Pool)



(c) $k = 3$ (after FC1)



(d) $k =$ random

Figure 5: Training and validation loss for Manifold Mixup at each layer setting ($\alpha = 0.2$).

Observations:

- Mixing at the earliest hidden layer ($k = 1$) gives the best accuracy (0.7522), comparable to input-space Mixup ($\alpha = 0.4$: 0.7556). Earlier mixing acts closer to input Mixup, preserving low-level feature diversity.
- Mixing at deeper layers ($k = 2$, $k = 3$) reduces accuracy slightly — at those stages, feature maps are more task-specific, so interpolation in that space is less natural and may produce out-of-manifold representations.
- Random k per batch performs worst among the four, likely because inconsistent mixing depth prevents the model from adapting its intermediate representations stably.
- All Manifold Mixup variants outperform traditional augmentation (0.7214) and are competitive with raw training (0.7392).